

The Diagonal Barrier in Kummer Large-Sieve Transfers

Jared Wilder

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Abstract

Paper 4 formulated a rank-two Hecke large-sieve bridge for arbitrary coefficient weights on Kummer vectors. This note proves that such a bridge is false as stated. The finite vector space $\mathbb{F}_\ell^2$ imposes an exact Plancherel identity with an unavoidable diagonal and fiber-sum term. A single localized coefficient already contributes $\ell^2 - 1$, contradicting any bound of size $\ell^2(\log Q)^{-A}$ for arbitrary unit L^2 weights. The correct finite object is the vector-fiber variance, not a universal arbitrary-coefficient large sieve. Projective slope collision control does not imply vector-fiber control. The Bernoulli-Kummer smoothing theorem supplies a distributional version for the obstruction law, but Kummer-BV/BVK still require a separate sieve-transfer estimate for the actual von Mangoldt or sieve weights.

1 Setup

Let

$$G = \mathbb{F}_\ell^2.$$

For each prime $q \in \mathcal{Q}$, let

$$x_q = (\log_q 2, \log_q 3) \in G.$$

For complex weights ω_q , define the Fourier sums

$$S_A(\omega) = \sum_{q \in \mathcal{Q}} \omega_q e_\ell(A \cdot x_q), \quad A \in G,$$

and the vector-fiber sums

$$\Omega(v) = \sum_{q: x_q = v} \omega_q, \quad v \in G.$$

2 Exact identity

Theorem 1 (Exact Kummer large-sieve identity). *For arbitrary complex weights ω_q ,*

$$\sum_{A \neq 0} |S_A(\omega)|^2 = \ell^2 \sum_{v \in G} |\Omega(v)|^2 - \left| \sum_{q \in \mathcal{Q}} \omega_q \right|^2.$$

Proof. Since $S_A(\omega) = \sum_{v \in G} \Omega(v) e_\ell(A \cdot v)$, Plancherel on G gives

$$\sum_{A \in G} |S_A(\omega)|^2 = \ell^2 \sum_{v \in G} |\Omega(v)|^2.$$

The $A = 0$ term is

$$|S_0(\omega)|^2 = \left| \sum_q \omega_q \right|^2.$$

Subtracting it proves the identity. □

Corollary 1 (Diagonal counterexample). *The bound*

$$\sum_{A \neq 0} |S_A(\omega)|^2 \ll_A \ell^2 (\log Q)^{-A}$$

cannot hold for all weights satisfying $\sum_q |\omega_q|^2 \leq 1$.

Proof. Choose one prime q_0 and set $\omega_{q_0} = 1$, $\omega_q = 0$ for $q \neq q_0$. Then

$$|S_A(\omega)|^2 = 1 \quad (A \neq 0),$$

so

$$\sum_{A \neq 0} |S_A(\omega)|^2 = \ell^2 - 1.$$

For fixed ℓ , the proposed right-hand side tends to 0 relative to ℓ^2 as $Q \rightarrow \infty$. In the high- ℓ range it is still smaller by an arbitrary logarithmic power. The stated arbitrary-coefficient saving is therefore false. □

Corollary 2 (Fiber multiplicity barrier). *If M indices q share the same vector $x_q = v_0$, and $\omega_q = M^{-1/2}$ on those indices and 0 elsewhere, then $\sum_q |\omega_q|^2 = 1$ but*

$$\sum_{A \neq 0} |S_A(\omega)|^2 = M(\ell^2 - 1).$$

Thus the operator norm is controlled by vector-fiber multiplicity, not only by the ambient dimension.

Proof. Here $\Omega(v_0) = M^{1/2}$, all other fibers vanish, and $\sum_q \omega_q = M^{1/2}$. Substitution into [Theorem 1](#) gives

$$\ell^2 M - M = M(\ell^2 - 1).$$

□

3 Distributional version

The identity becomes useful when the weights define a probability distribution rather than arbitrary localized coefficients.

Proposition 1 (Probability-weight Plancherel). *Let*

$$\mu = \sum_{q \in \mathcal{Q}} a_q \delta_{x_q}, \quad a_q \geq 0, \quad \sum_q a_q = 1.$$

Then

$$\sum_{A \neq 0} |\widehat{\mu}(A)|^2 = \ell^2 \|\mu - u\|_2^2,$$

where u is uniform on G .

Proof. Apply [Theorem 1](#) with $\omega_q = a_q$. The fiber sums are the probabilities $\mu(v)$, and the $A = 0$ term is 1. Since

$$\|\mu - u\|_2^2 = \sum_v \mu(v)^2 - \ell^{-2},$$

we get

$$\ell^2 \sum_v \mu(v)^2 - 1 = \ell^2 \|\mu - u\|_2^2.$$

□

Corollary 3 (Bernoulli-Kummer distributional large sieve). *Let ν be the Bernoulli-Kummer obstruction law from Paper 12. If*

$$W_\ell(\mathcal{Q}) = \min_{A \neq 0} \sum_{q: A \cdot x_q \neq 0} p_q(1 - p_q),$$

then

$$\sum_{A \neq 0} |\widehat{\nu}(A)|^2 \leq (\ell^2 - 1) \exp\left(-\frac{16}{\ell^2} W_\ell(\mathcal{Q})\right).$$

In particular, if $16W_\ell(\mathcal{Q})/\ell^2 \geq A \log \log Q + O(\log \ell)$, then

$$\sum_{A \neq 0} |\widehat{\nu}(A)|^2 \ll_A (\log Q)^{-A}.$$

Proof. Paper 12 proves

$$\|\nu - u\|_2^2 \leq (1 - \ell^{-2}) \exp\left(-\frac{16}{\ell^2} W_\ell(\mathcal{Q})\right).$$

Multiply by ℓ^2 and use [Proposition 1](#).

□

4 Projective slope control is not vector-fiber control

The projective Kummer collision graph records the class

$$\sigma_\ell(q) = [x_q] \in \mathbb{P}^1(\mathbb{F}_\ell)$$

when $x_q \neq 0$. This forgets the radial scalar in x_q . Additive characters $e_\ell(A \cdot x_q)$, however, see the full vector.

Proposition 2 (Projective-to-vector gap). *Small projective slope collision does not imply small vector-fiber variance for arbitrary coefficients.*

Proof. Choose one nonzero vector on each projective line in $\mathbb{F}_\ell^2$, and put equal probability mass on these $\ell + 1$ representatives. The projective slope distribution is perfectly uniform. But the vector distribution has support size only $\ell + 1$, so

$$\|\mu - u\|_2^2 = \frac{1}{\ell + 1} - \frac{1}{\ell^2},$$

which is much larger than a logarithmic power saving target in the high- ℓ range. By [Proposition 1](#), the corresponding nontrivial character energy is

$$\ell^2 \left(\frac{1}{\ell + 1} - \frac{1}{\ell^2} \right) \asymp \ell.$$

Thus slope equidistribution alone does not give vector Fourier saving. Arbitrary coefficients localized on a single vector fiber give the stronger obstruction in [Corollaries 1 and 2](#).

□

5 Corrected sieve-transfer status

The finite correction is:

- There is no arbitrary-coefficient Kummer large-sieve saving of size $\ell^2(\log Q)^{-A}$ under only $\sum |\omega_q|^2 \leq 1$.
- The exact finite quantity is the vector-fiber energy $\sum_v |\Omega(v)|^2$.
- Projective KRC is a slope-statistic; it does not control radial vector fibers.
- Bernoulli-Kummer smoothing supplies a distributional large-sieve statement for the obstruction probability law.
- Kummer-BV and BVK still require a separate analytic transfer statement for the actual von Mangoldt or sieve weights. That transfer must control vector-fiber sums for those weights, or state an equivalent Fourier variance estimate directly.

Warning 1 (What this does not prove). *This note does not prove Kummer-BV, BVK, or Erdős–Graham #203. It proves that one proposed arbitrary-coefficient bridge is false and replaces it with the exact finite identity that any valid sieve-transfer theorem must satisfy.*